

JACKSON HULL

Madison, WI | Jackson@hulitech.com
linkedin.com/in/jackson-hull-132966278 | github.com/strawcabbage

EDUCATION

- University of Wisconsin-Madison**, Madison, WI Expected May 2028
B.S. Computer Science and Physics | GPA: 3.5/4.0 | Dean's List
Relevant Coursework: Distributed Systems and Big Data, Machine Organization and Programming, Data Structures and Algorithms
Activities: Wisconsin AI Safety Initiative; Quantum Computing Club
Industry Liaison, Wisconsin Robotics: Secured ~\$30K in sponsorships, adding Mastermold and Oshkosh Corporation as new sponsors and renewing Onshape and Protocase.
- University of Minnesota**, Quantum + Chips Program Summer 2025
One of 25 selected nationwide for a quantum computing and chip design program with NVIDIA, Google, Intel.

EXPERIENCE

- AI Engineer Intern**, T-Mobile (IntentCX Team) | *Python, Databricks, MongoDB, Splunk, W&B Weave* May 2026 to Aug 2026
- Led the design of a CI/CD regression framework that catches voice agent regressions before they reach customers, partnering with QA and eval teams and writing the design doc and technical documentation through cross-functional review; projected to save \$200K+ annually across QA, hotfix, and incident costs.
 - Shipped AI-assisted tooling, adopted team-wide, that reproduces Jira defects via the customer-facing voicebot at triage and post-fix verification, automating away 90%+ of a 15-minute manual step across ~90 tickets per release (~3,400+ engineer-hours annually).
 - Built eval cases from live production calls mined through Weave and Splunk into Databricks and MongoDB, then replayed them through scripted inputs and an LLM user simulator whose goal-driven personas let conversations evolve the way real callers do.
 - Graded each run with deterministic checks and tool-call assertions wherever behavior was predictable, reserving a pairwise LLM judge for open-ended turns and comparing against production across repeated order-swapped runs to reduce position bias.
- Undergraduate Research Assistant**, Distributed Systems, UW-Madison | *Python* Jan 2026 to Present
- Co-authoring a paper on LLM-generated scheduling policies under Prof. Tyler Caraza-Harter, covering estimator co-design, robustness to estimation error, and whether models should generate decision-making code or decide directly.
 - Built an agentic coding harness in Python that validates AI-generated code via simulation and LLM-guided evolutionary search, scaling distributed runs across 16 machines to ~4 s adjusted latency (~11 s on 4), ~70% faster than Codex and ahead of OpenCode.
- Undergraduate Research Assistant**, Data Infrastructure, Plourde Lab, UW-Madison | *Python, Kafka, SQLite* Sep 2025 to May 2026
- Owned a Kafka ingestion pipeline and SQLite storage layer end to end on Linux with on-call troubleshooting, keeping ~1.5M events per day reliable across 18 channels and 4 sensors in production for 6 months to track fabrication equipment health and device noise.
 - Debugged a production data-gap incident by isolating it to the sensor-to-Kafka link and tracing it to a failing USB cable, then added a staleness check that flags missing data in the UI so dropouts surface immediately instead of looking like a flat reading.
 - Replaced the lab's live-only instrument displays with real-time monitoring backed by historical queries and rolling time windows, surfacing a ~270M-event record for 6 researchers who previously could not reach it.
- Software Engineering Intern**, Paladin Cloud | *Java* Jun 2024 to Sep 2024
- Generated, code-reviewed, and validated 10 AI-generated cloud security policy test suites before deployment, with estimated savings of \$72 to \$144 per policy, and presented findings to engineering leadership, driving adoption into production workflows.

PROJECTS

- Unix Shell** *C, Linux*
- Built an interactive Unix shell in C with a custom tokenizer and parser supporting pipelines, I/O redirection, built-ins, and background job control, using SIGINT and SIGCHLD handlers to manage and reap child processes.
- Transformer Engine, Autograd and Inference Runtime** *C++17, SIMD, OpenMP, CMake, GoogleTest*
- Building a C++ tensor library with object-oriented design and reverse-mode autograd, hand-deriving backward passes for matmul, softmax, layer norm, and GELU, verified against finite-difference gradient checks under GoogleTest.
 - Adding SIMD matmul kernels and a KV cache for GPT-2 small inference, validated against a PyTorch reference implementation.
- BookReviews API, Scalable Backend Service** *Java, Spring Boot, PostgreSQL, Docker, AWS*
- Containerized a Spring Boot microservice with 23 REST endpoints and deployed it to AWS Elastic Beanstalk on PostgreSQL RDS with Flyway-managed schema migrations, benchmarking it at 9.8 ms p50 and 30.5 ms p95 latency over 500 requests.
 - Backed the reward system with transactional writes so that a review and the points it awards commit or roll back as a single unit, keeping user balances consistent across concurrent requests.

TECHNICAL SKILLS

Languages: C++ | Rust | Go | Python | C | Java
Systems and Infrastructure: Linux | TCP/IP | Docker | AWS | CI/CD | Git
Backend and Data: Apache Kafka | PostgreSQL | MongoDB (NoSQL) | SQLite | Spring Boot | Databricks
Observability and ML: Splunk | W&B Weave | PyTorch | LangGraph | Claude Code